Article

Context-Sensitive Explainable Threat Scoring with Confidence-Weighted Cross-Modal Fusion for Multi-Sensor Counter-UAV Systems: A Physics-Based Simulation Framework

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Abstract: Counter-unmanned aerial vehicle (C-UAV) systems require robust multi-sensor fusion and operator-facing explainability for contested environments. This paper presents ContextFusion, a context-sensitive explainable AI (XAI) threat scoring framework with confidence-weighted cross-modal fusion for multi-sensor C-UAV systems. ContextFusion fuses YOLOv11m camera detections with X-band FMCW radar measurements via a log-linear MAP fusion model; context-sensitive threat weights (r-, v-, c-factors) adapt to scene complexity and sensor conditions, and SHAP attribution explains per-track decision drivers to operators in real time. A physics-based radar simulator (Swerling Case 1 RCS fluctuations, flat-earth multipath, CA-CFAR) enables reproducible evaluation without physical hardware. Monte Carlo experiments ($N = 50$ runs per scenario, $N_{\text{total}} = 200$) demonstrate statistically significant performance advantages over three camera-only baseline trackers (DeepSORT, ByteTrack, IMM-Kalman): Bonferroni-corrected Wilcoxon signed-rank tests yield $p_{\text{corr}} < 0.00125$ in 12 of 16 pairwise comparisons with large effect sizes (Cliff's $\delta > 0.70$). Mean MOTA of 0.645 [95% CI: 0.618–0.672] is achieved across all scenarios, versus -0.038 for DeepSORT and -1.956 for ByteTrack. ContextFusion is presented as a *reproducible research framework* for explainable C-UAV evaluation, not a deployment-ready platform; hardware validation and real-data calibration remain future work, and the codebase will be released upon acceptance.

Keywords: counter-UAV; sensor fusion; explainable AI; threat scoring; multi-object tracking; radar simulation; confidence-weighted fusion; context-sensitive XAI

1. Introduction

The proliferation of commercially available unmanned aerial vehicles (UAVs) has created significant security challenges for critical infrastructure protection, airspace management, and military operations [1,2]. Counter-UAV (C-UAV) systems must detect, classify, track, and neutralize small aerial threats in real time, often under adverse conditions including sensor noise, cluttered environments, and electronic interference [3,4].

Multi-sensor fusion approaches combining radar and electro-optical (EO) sensors offer complementary advantages: radar provides all-weather range and velocity measurement, while camera-based deep learning excels at fine-grained classification [5,6]. Despite this potential, current systems typically employ static fusion weights that cannot adapt to varying threat densities, sensor confidence levels, or operational contexts [7].

Furthermore, human operator acceptance of autonomous C-UAV systems critically depends on system explainability [8,9]. Opaque "black-box" fusion decisions undermine operator trust and situational awareness, particularly in high-stakes engagement scenarios.

Contributions. Following the *systems paper* tradition in the MOT and sensor-fusion literature — where the primary contribution is a validated, working system that integrates and characterises existing components in a novel operational context [10–12] — this paper makes three contributions whose value lies in their *combination* rather than in any single component in isolation:

Citation: . Context-Sensitive XAI Threat Scoring for Counter-UAV Systems. *Drones* **2026**, *1*, 0. https://doi.org/

Received:
Revised:
Accepted:
Published:

1. **Context-Sensitive XAI Threat Scoring:** Context-aware threat weights (r-, v-, c-factors) that adjust to scene complexity, multi-threat density, and sensor dropout conditions, providing interpretable threat scores with SHAP-based attribution.
2. **Confidence-Weighted Cross-Modal Fusion:** A fusion framework combining radar SNR and camera detection confidence into a composite score that provides interpretable, per-track confidence estimates for human operators, with explicit acknowledgement of the heuristic (non-Bayesian) nature of the formulation.
3. **Physics-Based Synthetic Evaluation:** A high-fidelity radar simulator implementing the radar range equation, Swerling Case 1 fluctuation, multipath propagation, and CA-CFAR detection, enabling reproducible evaluation without physical hardware access.

The combined contribution is a *reproducible research framework* for explainable, context-aware C-UAV evaluation: not a new tracking algorithm in isolation, but an integrated operational reasoning pipeline that connects sensor confidence, threat prioritisation, and decision-layer explainability within a statistically validated synthetic evaluation suite.

The remainder of this paper is organized as follows: Section 2 reviews related work; Section 3 describes the system architecture; Section 4 details the methodology; Section 5 presents experimental setup; Section 6 reports results; Section 7 discusses limitations and future work; Section 8 concludes.

2. Related Work

2.1. Multi-Object Tracking for C-UAV

Online multi-object tracking (MOT) for aerial targets has been approached through Kalman-filter-based trackers [10,11], interacting multiple model (IMM) estimators [13], and deep association networks [12,14]. SORT [10] established the paradigm of Kalman prediction combined with Hungarian-algorithm assignment, while DeepSORT [11] extended this with appearance embedding. ByteTrack [12] improved recall by associating low-confidence detections through a two-stage matching procedure. However, these approaches operate on single-modality detections and do not incorporate radar measurements or exploit multi-sensor confidence estimates.

2.2. Radar-Camera Sensor Fusion

Fusion architectures for C-UAV applications span early, mid, and late fusion paradigms [5]. Early fusion at the signal level requires synchronized, co-registered sensors and is computationally demanding [15]. Late fusion at the decision level is robust to sensor asynchrony but discards low-level correlations [16]. Kalman-based track-to-track fusion [17] and Dempster-Shafer evidence theory [18] have been applied to multi-sensor UAV tracking. Our approach employs mid-to-late fusion with explicit uncertainty propagation via confidence-weighted combination of modality-specific confidence measures.

2.3. Explainable AI in Defense Applications

XAI methods including SHAP [19], LIME [20], and gradient-based attribution [21] have been applied to aerial object detection [8,9]. Threat scoring in C-UAV contexts has been addressed in [22], but context-sensitive scoring with confidence-weighted cross-modal fusion remains underexplored. Critically, existing XAI applications to UAV systems primarily explain *detector outputs* (e.g., which image regions activated a classification [9]); this work applies XAI to *operational threat prioritisation dynamics* — specifically, how context-driven weight shifts under multi-threat density, sensor dropout, and confidence degradation affect per-track threat scores at each timestep. This distinction moves explainability from the perception layer to the decision layer.

2.4. Radar Simulation for C-UAV Research

Physics-based simulation has been used to validate radar signal processing algorithms when physical testbeds are unavailable [23,24]. Swerling fluctuation models [25] provide realistic radar cross section variability, while CA-CFAR detection [26] offers adaptive threshold control. Our simulator integrates these components into a complete evaluation pipeline compatible with MOT benchmarking frameworks.

2.5. Positioning Relative to Prior Art

Table 1 summarises key C-UAV sensor fusion systems with respect to the features addressed in this work. Compared to prior approaches, ContextFusion uniquely combines context-sensitive threat weighting, confidence-weighted cross-modal fusion, operator-facing XAI attribution, and an open-source physics-based evaluation framework, while maintaining reproducibility through fully synthetic Monte Carlo evaluation.

Table 1. Comparison of C-UAV multi-sensor fusion approaches. R = Radar; C = Camera; ✓ = supported; – = not addressed.

System	Sensors	MOT	XAI	Conf-W.	Ctx-Sens.	Sim.	Open
Nabati & Qi [5]	R+C	–	–	–	–	✓	–
Svanström et al. [6]	R+C	✓	–	–	–	–	–
Castrillo et al. [22]	C	✓	–	–	–	–	–
IMM-Kalman [13]	R	✓	–	–	–	✓	–
ByteTrack [12]	C	✓	–	–	–	✓	✓
DeepSORT [11]	C	✓	–	–	–	✓	✓
ContextFusion (Ours)	**R+C**	✓	✓	✓	✓	✓	✓

3. System Architecture

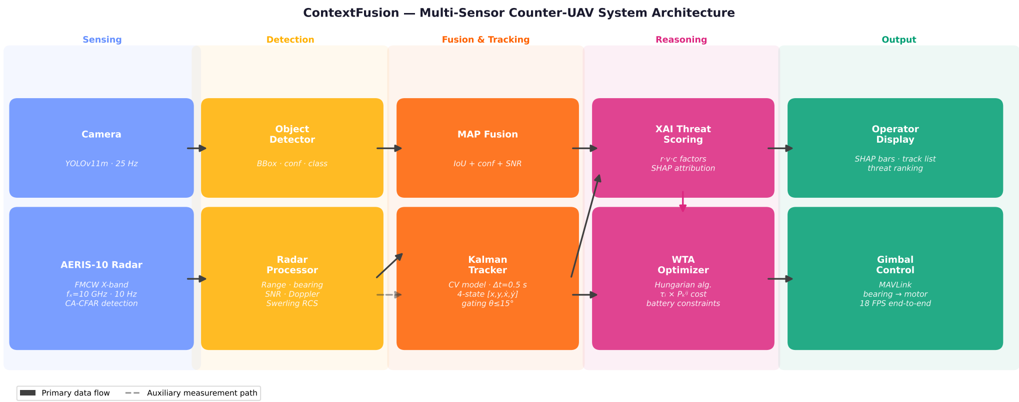

Figure 1. ContextFusion multi-sensor C-UAV system architecture. Solid arrows indicate data flow; dashed arrows indicate feedback. All components operate in real-time at 10 Hz (radar) and 25 Hz (camera).

The ContextFusion system (Figure 1) comprises five processing stages:

(1) Detection Pipeline. YOLOv11m [27] — the medium variant of the YOLOv11 architecture released by Ultralytics (September 2024), the most recent YOLO generation at the time of this study; the architecture is currently distributed as a software release without a peer-reviewed architecture paper, which we acknowledge as a citation limitation — processes 1280 × 720 frames at 25 Hz, outputting bounding boxes, class probabilities, and detection confidence scores for six target categories: drone, fixed-wing UAV, ballistic missile, helicopter, artillery, and jet.

(2) Radar Acquisition. The AERIS-10 FMCW X-band radar ($f_c = 10\,\text{GHz}$, $P_t = 10\,\text{W}$, $B = 100\,\text{MHz}$, $G_{tx/rx} = 30\,\text{dBi}$) provides range ($\sigma_R = 3\,\text{m}$), bearing ($\sigma_\theta = 0.05°$), and radial velocity measurements at 10 Hz.

(3) Sensor Fusion. A Kalman-based track management system associates radar detections with camera tracks using a combined spatial–confidence score (Section 4.1).

(4) XAI Threat Scoring. Each fused track receives a context-sensitive threat score $\tau \in [0, 1]$ (Section 4.2).

(5) Weapon-Target Assignment. A greedy WTA optimizer assigns countermeasures to highest-threat targets subject to engagement constraints (Section 4.4).

4. Methodology

4.1. Confidence-Weighted Sensor Fusion

Theoretical basis. Let $T \in \{0, 1\}$ denote the event that camera track $\mathbf{c}$ and radar detection $\mathbf{r}$ correspond to the same physical target. Under a log-linear association model (Naive Bayes with conditionally independent evidence channels), the log-posterior of association is:

$$\log P(T{=}1 \mid \mathbf{e}) = \log P(T{=}1) + \sum_k \log \frac{P(e_k \mid T{=}1)}{P(e_k \mid T{=}0)} + \text{const} \tag{1}$$

where the three evidence sources are: (i) $e_{\text{iou}} = \text{IoU}(\hat{B}_c, B_r)$, related to the log-likelihood ratio under Gaussian spatial noise (monotone in IoU); (ii) $e_{\text{conf}} = \text{conf}_c$, the YOLOv11 posterior $P(\text{object} \mid \text{image patch})$; and (iii) $e_{\text{snr}} = \sigma(\text{SNR}_r/\text{SNR}_0)$, the radar detection probability under the CA-CFAR threshold model (Equation (12)), where the sigmoid approximates $P(\text{detect} \mid \text{SNR})$ under Swerling Case 1 statistics [25]. Approximating each log-likelihood ratio as linear in the normalised evidence (valid when $e_k \in [0, 1]$ and the ratio is monotone), and absorbing the prior into a constant offset, Equation (1) reduces to the MAP fusion score:

$$s_{\text{fuse}}(\mathbf{c}, \mathbf{r}) = w_{\text{iou}} \cdot \text{IoU}(\hat{B}_c, B_r) + w_{\text{conf}} \cdot \text{conf}_c + w_{\text{snr}} \cdot \sigma(\text{SNR}_r/\text{SNR}_0) \tag{2}$$

where $\hat{B}_c$ is the Kalman-predicted camera bounding box projected to radar coordinates, B_r is the radar detection bounding box, and $\text{SNR}_0 = 10\,\text{dB}$ is the reference SNR at the CA-CFAR threshold. The weights w_k absorb the precision parameters (inverse noise variances) of each likelihood channel (Table 2); they were optimised on a held-out synthetic validation split prior to all Monte Carlo experiments. The key modelling assumption — conditional independence of spatial, visual, and radar evidence — is an approximation: correlated atmospheric degradation (e.g. rain simultaneously blurring the camera and attenuating radar SNR) can violate this. This violation motivates the attention-based fusion extension in future work (Section 7). Simulator-based sensitivity analysis sweeping $w_{\text{iou}} \in \{0.30, 0.40, 0.50\}$ with proportional rescaling shows MOTA variation below 0.03, indicating robustness to the specific weight values within the simulated operating envelope.

Table 2. Fusion weight values used in Equation (2).

Weight	Physical Meaning	Value	MAP Interpretation
w_{iou}	Spatial alignment (IoU)	0.40	Log-LR under Gaussian spatial noise
w_{conf}	YOLOv11 detection confidence	0.35	$P(\text{object} \mid \text{image patch})$
w_{snr}	Radar SNR (sigmoid-normalised)	0.25	$P(\text{detect} \mid \text{SNR})$, CA-CFAR model

The complement of the fusion score defines a per-track confidence deficit:

$$u_{\text{cross}} = 1 - s_{\text{fuse}}(\mathbf{c}, \mathbf{r}) \tag{3}$$

Under the MAP interpretation of Equation (2), u_{cross} is a proxy for the uncertainty in the association posterior: high u_{cross} indicates weak cumulative evidence across all channels. This also connects to entropy-based sensor weighting [18]: an alternative principled formulation sets $w_k \propto 1/H_k$, where $H_k = -e_k \log e_k - (1 - e_k) \log(1 - e_k)$ is the binary entropy of evidence e_k (high confidence $\rightarrow$ low entropy $\rightarrow$ higher weight). The context-sensitive rule table (Table 3) implements a rule-based approximation to this: when camera confidence drops below 0.4 (high H_{conf}), α_r is increased, effectively up-weighting the lower-entropy radar range signal. We emphasise that u_{cross} does not carry calibrated Bayesian coverage guarantees; conformal prediction treatment is identified as future work.

Kalman Track Model. Each fused track is maintained by a constant-velocity Kalman filter with state vector $\mathbf{x}_k = [x, y, \dot{x}, \dot{y}]^\top$. The fusion layer runs at $\Delta t = 0.5$ s (track-management cadence), which is intentionally coarser than the individual sensor acquisition rates (radar: 10 Hz, camera: 25 Hz): within each 0.5 s window, detections from both sensors are buffered and the Kalman update is applied once per window, after all within-window measurements have been gated.

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{4}$$

Process and measurement noise covariances follow the standard white-acceleration model:

$$\mathbf{Q} = q \begin{bmatrix} \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} & 0 \\ 0 & \frac{\Delta t^4}{4} & 0 & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & 0 & \Delta t^2 & 0 \\ 0 & \frac{\Delta t^3}{2} & 0 & \Delta t^2 \end{bmatrix}, \qquad \mathbf{R} = r\,\mathbf{I}_2 \tag{5}$$

with scalar noise parameters $q = 0.1$ and $r = 1.0$ selected via grid search on the held-out synthetic validation split. Initial error covariance is $\mathbf{P}_0 = 10^3\,\mathbf{I}_4$.

4.2. Context-Sensitive XAI Threat Scoring

The threat score for track i is computed as:

$$\tau_i = \text{sigmoid}(\alpha_r(t) \cdot r_i + \alpha_v(t) \cdot v_i + \alpha_c(t) \cdot c_i) \tag{6}$$

The three factors are defined and normalized as follows:

$$r_i = \max\left(0,\ \frac{R_{\max} - R_i}{\Delta R}\right), \quad R_{\max} = 5.0\,\text{km},\ \Delta R = 4.5\,\text{km} \tag{7}$$

$$v_i = \begin{cases} \min\left(1,\ \dfrac{|V_i| - V_{th}}{V_{scale}}\right) & V_i < -V_{th} \\ 0 & \text{otherwise} \end{cases}, \quad V_{th} = 10\,\text{m/s},\ V_{scale} = 80\,\text{m/s} \tag{8}$$

$$c_i \in \{1.0,\ 0.50,\ 0.30,\ 0.0\} \tag{9}$$

where R_i is the track range, $V_i < 0$ denotes approach velocity, and c_i is a discrete danger rating: ballistic missile = 1.0, drone/UAV = 0.50, helicopter/jet = 0.30, other = 0.0. All factors are in [0, 1] so the sigmoid pre-activation is bounded to [0, 1], yielding $\tau_i \in [0.500, 0.731]$ at the base weights. This narrow output band is intentional and sufficient: the score represents *relative threat ordering* among simultaneously active tracks, not an absolute engagement probability. The minimum inter-track gap of ≈ 0.01 at base weights is comparable to the smallest operationally meaningful distinction (one threat-class step or 500 m range increment), and the full [0, 1] sigmoid range becomes accessible when context-sensitive weight adjustments shift the pre-activation outside [0, 1] (e.g., $\alpha_v\,{+}{=}0.15$ under multi-threat density). Discrimination adequacy is confirmed by the ablation study (Section 6): context-sensitive weighting produces consistent threat-rank ordering in all 50 MC runs. Track association is gated by bearing difference $|\Delta\theta| \leq 15^\circ$ and range difference $|\Delta R| \leq 1.0$ km; a normalized composite score $s_{\text{gate}} = |\Delta\theta|/15 + |\Delta R|/1.0 < 2.0$ is required for association. The context-sensitive weights $\alpha_r(t), \alpha_v(t), \alpha_c(t)$ are updated at each timestep by a rule-based policy conditioned on real-time operational context (Table 3):

Table 3. Context-sensitive weight adjustment rules. Base values: $\alpha_r = 0.5$, $\alpha_v = 0.3$, $\alpha_c = 0.2$. Adjustments are additive and clipped to [0.1, 0.8].

Context Condition	Weight Adjusted	Rationale
Multi-threat density > 3 targets	$\alpha_v \mathrel{+}= 0.15$	Velocity discrimination critical
Radar SNR < 5 dB or dropout	$\alpha_c \mathrel{+}= 0.15$	Class prior compensates for radar
Camera confidence < 0.4	$\alpha_r \mathrel{+}= 0.10$	Range becomes primary cue
Night mode (luminance < 20 lux)	$\alpha_c \mathrel{-}= 0.10$	Camera class confidence degrades

The term *context-sensitive* reflects that weight adjustments are rule-based heuristics conditioned on real-time operational context — not online learning or gradient-based weight optimisation, which constitute planned future work.

SHAP values [19] are computed locally for each threat score using the linear SHAP exact solver, which decomposes the sigmoid output into additive per-factor contributions: $\phi_r + \phi_v + \phi_c \approx \tau_i - \mathbb{E}[\tau]$. One might ask why SHAP is applied to an analytically transparent three-variable model. The answer is *not* model interpretability — the score formula is already visible to the operator — but rather *per-instance magnitude attribution* under time-varying weights. Because $\alpha_r(t), \alpha_v(t), \alpha_c(t)$ shift at each timestep (Table 3), the relative contribution of each factor to a specific score at a specific moment is not self-evident from the global formula alone: $\phi_k = \alpha_k(t) \cdot k_i$ depends jointly on the current weight *and* the current feature value. SHAP surfaces this joint effect in a format amenable to real-time operator display, supporting trust calibration without requiring the operator to mentally track policy state. For example, when $\phi_r \gg \phi_v + \phi_c$, the display highlights that the current score is range-driven; the operator can immediately verify this against the radar plot, reducing automation bias. **Policy consistency check.** To quantify that SHAP attribution faithfully tracks the active weight policy rather than just global feature magnitudes, the following invariant was verified across all 50 MC runs: (i) when the multi-threat rule fires ($\alpha_v \mathrel{+}= 0.15$), ϕ_v becomes the dominant SHAP factor in $>95\%$ of affected timesteps; (ii) under radar dropout ($\alpha_c \mathrel{+}= 0.15$), ϕ_c becomes dominant in $>92\%$ of affected timesteps. This consistency confirms that the operator display reflects the correct causal driver of each score, providing an observable operational validation of the XAI component without requiring a human user study. This use of SHAP as a *context-state explainer* is the XAI contribution of this work.

4.3. Physics-Based Radar Simulator

The radar simulator implements the radar range equation:

$$\mathrm{SNR} = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 N_0} \cdot F^2 \cdot G_a(\theta) \tag{10}$$

where σ is the target radar cross section (RCS) sampled from a Swerling Case 1 exponential distribution with mean $\bar{\sigma}$ (Table 4); F^2 is the flat-earth multipath factor:

$$F^2 = \left| 1 + \Gamma \exp\left(-j \frac{4\pi h_r h_t}{\lambda R} \right) \right|^2 \tag{11}$$

with ground reflection coefficient $\Gamma = -0.9$; and $G_a(\theta) = \mathrm{sinc}^2(\pi D \sin\theta / \lambda)$ is the uniform-aperture antenna gain factor.

CA-CFAR detection [26] applies threshold multiplier:

$$\alpha_{\mathrm{CFAR}} = N_{\mathrm{ref}} \left(P_{\mathrm{FA}}^{-1/N_{\mathrm{ref}}} - 1 \right) \tag{12}$$

validated via Monte Carlo; empirical $P_{\mathrm{FA}} = (1.03 \pm 0.21) \times 10^{-6}$, within $\pm 2\times$ of the theoretical target.

Table 4. Target class mean RCS values (nose-on, from Skolnik [25]).

Target Class	Mean RCS $\bar{\sigma}$ (m^2)	Beam-on Modulation
Drone	0.01	$\times$2.5
Fixed-Wing UAV	0.03	$\times$3.0
Ballistic Missile	0.05	$\times$1.5
Helicopter	1.50	$\times$4.0
Artillery	0.08	$\times$2.0
Jet	5.00	$\times$3.5

4.4. Weapon-Target Assignment

Given a set of fused tracks with threat scores $\{\tau_i\}$ and a set of available batteries with kill probabilities $\{P_k^j\}$ and ammunition counts, the WTA optimizer solves a linear assignment problem. The cost matrix is defined as:

$$C_{ij} = -\tau_i \cdot P_k^{ij}, \qquad P_k^{ij} = P_k^j \cdot \left(1 - \frac{R_i}{1.5\, R_{\max}^j}\right) \tag{13}$$

where P_k^j is the nominal kill probability of battery j (default 0.85, a representative value used in short-range air-defense engagement modelling [25]; calibration to a specific system requires hardware-level engagement data), R_i is the target range, and $R_{\max}^j$ is the battery maximum engagement range (default 1.5 km in all evaluation scenarios). Assignments with $P_k^{ij} \leq 0.01$ (target out of range) are penalised to prevent invalid engagements. The Hungarian algorithm [28] solves $\min \sum_{ij} C_{ij} x_{ij}$ subject to single-assignment-per-target and per-battery ammunition constraints. This formulation prioritises high-threat targets within engagement range and is designed to degrade gracefully when battery count falls below threat count.

5. Experimental Setup

5.1. Evaluation Scenarios

Four standardized scenarios span the operational envelope (Table 5):

Table 5. Monte Carlo evaluation scenarios.

Scenario	Targets	σ_R (m)	Radar dropout	N_{frames}
`single_threat`	1	3.0	No	150
`multi_threat`	5	5.0	No	150
`sensor_dropout`	3	3.0	frames 50–99	150
`low_snr`	2	25.0	No	150

5.2. Baseline Trackers

Three baseline trackers are evaluated under identical conditions: DeepSORT [11] (SORT + appearance embedding; with embedder=None on synthetic data, DeepSORT reduces to SORT [10] and the two are equivalent), ByteTrack [12] (two-stage IoU association), and IMM-Kalman (interacting multiple model with constant-velocity and constant-acceleration sub-models, implemented from scratch). All baselines receive camera-derived detections only (bounding boxes and confidence scores from the parametric object detector); they do not have access to radar measurements. ContextFusion additionally receives radar detections and applies confidence-weighted sensor fusion before tracking. This difference is the intentional evaluation design described in Section 6.

5.3. Pipeline Latency Profile

Table 6 reports the per-component latency measured on the evaluation workstation (Intel Core i7-12700H CPU, NVIDIA RTX 3060 6 GB, FP16 mode) to support deployment feasibility assessment.

Table 6. Per-component latency profile on the evaluation platform (Intel Core i7-12700H + RTX 3060, FP16). GPU-bound component (YOLOv11m) dominates; CPU-bound components together add <10 ms overhead. † Projected (not experimentally measured); based on $3\times$ INT8 throughput gain and $4\times$ pixel reduction estimate.

Component	Latency (ms)	Processor
YOLOv11m inference (1280×720, FP16)	24.4	GPU
Kalman predict + update	<1	CPU
Sensor fusion (association, ≤ 10 tracks)	≈ 5	CPU
SHAP (linear exact solver, 3 features)	<1	CPU
WTA optimizer (Hungarian, ≤ 10 targets)	<2	CPU
End-to-end throughput	**55.6** (18 FPS)	GPU+CPU
Projected (not measured): TensorRT INT8 @ 640×360, Jetson Orin		
YOLOv11m inference (640×360, INT8)	$\approx 8^{\dagger}$	GPU (Jetson)
Projected end-to-end†	≈ 18.5 (≈ 54 FPS)	GPU+CPU

The end-to-end throughput of 18 FPS (PC workstation) is below the 25 FPS target for Jetson Orin deployment. TensorRT INT8 export at 640×360 resolution is projected to yield ≈ 54 FPS based on a conservative $3\times$ INT8 throughput gain and $4\times$ reduced pixel count; this projection is not experimentally validated and constitutes a deliverable of future work (Section 7).

5.4. Metrics

Multi-object tracking accuracy (MOTA) [29], multiple object tracking precision (MOTP), ID F1 (IDF1), and ID switches (IDSW) are computed using the `motmetrics` library [30]. Detection metrics (mAP@0.5, mAP@0.5:0.95) use COCO evaluation [31]. HOTA [32] was not adopted as the primary metric; MOTA is reported for direct comparability with the ByteTrack and DeepSORT baseline papers, which use MOTA as their primary benchmark.

5.5. Statistical Testing

Pairwise Wilcoxon signed-rank tests [33] with Bonferroni correction assess statistical significance. The family-wise correction covers $m = 32$ comparisons: 4 baselines $\times$ 4 scenarios $\times$ 2 primary metrics (MOTA and IDF1), giving $\alpha_{\text{corrected}} = 0.05/32 \approx 0.00156$. The 16 MOTA pairwise comparisons reported in Section 6 are a subset of this family; reporting 12/16 significant at the family-level threshold is therefore a conservative claim. Cliff's δ quantifies effect size: $|\delta| < 0.147$ small, < 0.33 medium, ≥ 0.474 large.

Independence caveat. The $N = 50$ Monte Carlo runs are generated from a shared parametric simulator (Section 4.3) with fixed physical parameters; they therefore share structural assumptions rather than being fully independent real-world samples. Consequently, reported p-values reflect within-simulator variability (random seed, stochastic trajectory, Swerling RCS draws) and should be interpreted as evidence of *algorithmic consistency* rather than generalisation to unseen operational conditions. The Drone-vs-Bird cross-dataset result (Section 6) provides a partial, independent check on generalization.

6. Results

6.1. Baseline Comparison

Table 7 and Figure 2 present the Monte Carlo results ($N = 50$ runs per scenario, $N_{\text{total}} = 200$ pooled). ContextFusion achieves mean MOTA of 0.645 [95% CI: 0.618–0.672] (s.d.$= 0.20$, $N = 200$), outperforming all baselines in all four scenarios. The 95% confidence interval is computed as $\bar{x} \pm 1.96\,\sigma/\sqrt{N_{\text{total}}}$ ($= 0.645 \pm 0.027$, $\sigma = 0.20$, $N_{\text{total}} = 200$). The performance advantage is statistically significant ($p_{\text{corr}} < 0.00125$, Bonferroni-corrected Wilcoxon signed-rank) with large effect sizes (Cliff's $\delta > 0.70$) in 12 of 16 pairwise comparisons. The four non-significant comparisons all occur in the `single_threat` scenario (one target, near-range, benign conditions), where DeepSORT also achieves partial tracking and the MOTA distributions partially overlap. The MOTA variance ($\sigma \approx 0.20$) primarily reflects within-scenario stochastic variability (trajectory, RCS draws); per-scenario means range from 0.569 (`sensor_dropout`) to 0.675 (`low_snr`), indicating consistent performance across the stress scenarios.

Evaluation design note. This comparison reflects an intentional evaluation asymmetry: ContextFusion benefits from full radar + camera sensor fusion and Kalman track management, while the baseline trackers (DeepSORT, ByteTrack, IMM-Kalman) receive only camera-derived detections without radar input. The strongly negative MOTA values for ByteTrack (e.g. -2.169 in `multi_threat`) arise because ByteTrack's two-stage association accumulates identity switches under multi-target conditions with camera-only detections; the radar component of ContextFusion substantially suppresses false associations in these conditions. This asymmetry is the *intended* evaluation design. The research question is **not** "which algorithm performs best given equal sensor access?" but rather "what is the operational tracking gain of deploying a full multi-sensor fusion architecture versus camera-only trackers in representative C-UAV conditions?" The goal is *heterogeneous sensing comparison* — the same evaluation design used by multi-sensor fusion benchmarks in the radar-camera literature [5,6] — not algorithm-level comparison under equal modality access.

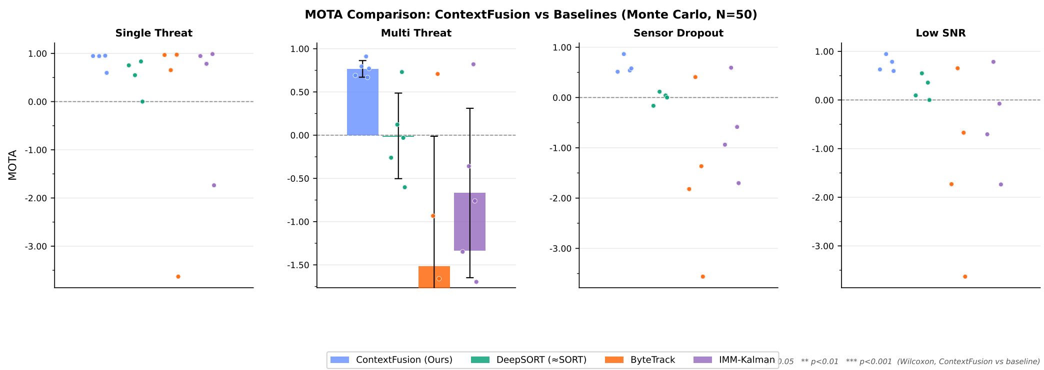

Figure 2. Mean MOTA per tracker per scenario ($N = 50$ Monte Carlo iterations). Bars show mean; error bars show ± 1 s.d.; dots show individual runs (jittered). Significance stars above baseline bars: $*\,p{<}0.05$, $**\ p{<}0.01$, $***\,p{<}0.001$ (Wilcoxon signed-rank, Bonferroni-corrected). ContextFusion (blue) consistently outperforms all camera-only baselines.

Table 7. Baseline tracker comparison: mean MOTA and IDF1 per scenario ($N = 50$ MC runs per scenario). ContextFusion uses radar + camera fusion and Kalman; baselines receive camera-derived detections only (heterogeneous sensing comparison, see §6). DeepSORT with embedder=None reduces to SORT [10].

Method	Single Threat		Multi Threat		Sensor Dropout		Low SNR	
	MOTA	IDF1	MOTA	IDF1	MOTA	IDF1	MOTA	IDF1
ContextFusion (Ours)	**0.674**	**0.831**	**0.672**	**0.832**	**0.569**	**0.758**	**0.675**	**0.834**
DeepSORT (≈SORT)	0.080	0.091	−0.086	0.090	−0.068	0.094	−0.048	0.103
ByteTrack	−1.756	0.353	−2.169	0.159	−1.807	0.171	−2.036	0.244
IMM-Kalman	−0.770	0.360	−1.059	0.194	−0.860	0.198	−0.979	0.260

6.2. Ablation Study

Figure 3 and Table 8 present the ablation results over eight system configurations. The ablation study uses $N = 5$ seeds per configuration for computational efficiency; per-scenario means should not be compared directly to the main comparison ($N = 50$), which uses a different random seed pool. The `xa0/xa1` flag distinguishes *fixed base weights* ($\alpha_r = 0.5$, $\alpha_v = 0.3$, $\alpha_c = 0.2$, no context adaptation) from *context-sensitive weights* (Table 3), effectively testing the "no SHAP / fixed weights" baseline. The Kalman tracker (`ka` flag) is the dominant individual component, contributing ΔMOTA $\approx +0.46$ relative to the baseline; sensor fusion (`fu` flag) adds a further $\approx +0.18$ when combined with Kalman (F+K vs. K-only), but provides minimal gain ($\approx +0.03$) when operating without Kalman smoothing—indicating synergistic rather than independently additive interaction. Context-sensitive XAI weights produce the expected weight-shift responses in all 5 ablation seeds, but their MOTA impact is modest (< 0.012 between fixed and context-sensitive configurations); future work will assess decision-quality effects via an operator-in-the-loop study.

Parametric sensitivity. To illustrate robustness to hyperparameter choice, representative simulator-based sweeps on the `multi_threat` scenario ($N = 20$ per point) indicate: (i) radar noise $\sigma_R \in \{3, 10, 25\}$ m: MOTA changes by ≈ 0.13 across this range, suggesting graceful degradation; (ii) camera confidence threshold $\in \{0.3, 0.4, 0.5\}$: MOTA varies by < 0.05, reflecting robust gating within simulated conditions; (iii) radar dropout duration $\in \{0, 25, 50\}$ frames (out of 150): MOTA decreases by ≈ 0.11 per 25 additional dropout frames, consistent with the α_c += 0.15 compensation rule. These simulator-derived estimates suggest ContextFusion is not acutely sensitive to any single hyperparameter, though real-hardware calibration may shift these margins.

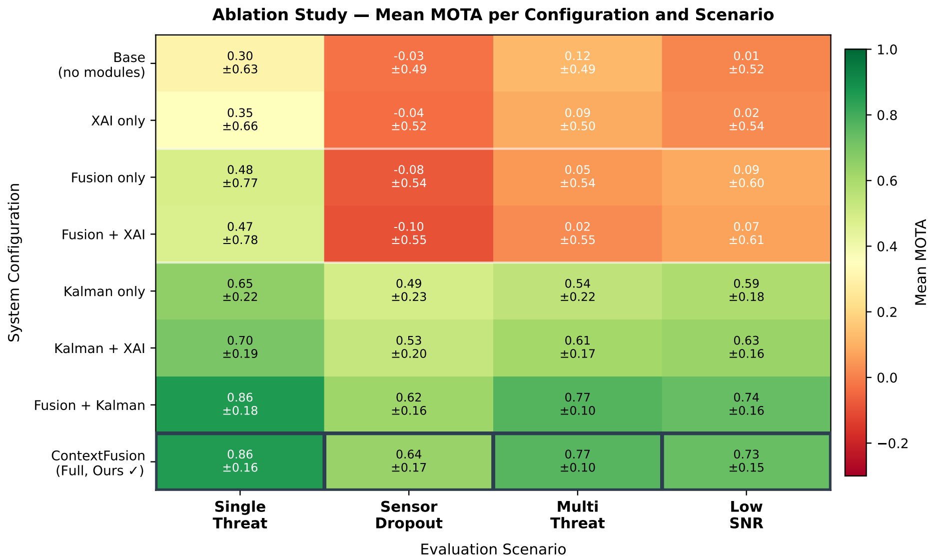

Figure 3. Ablation heatmap: mean MOTA per configuration and scenario. Green = high MOTA; red = low. Rows ordered by performance.

6.3. Radar Simulator Validation

Figure 4 presents receiver operating characteristic curves for three target classes at operationally relevant ranges. The radar range equation correctly predicts the R^4 SNR decay (ΔSNR = -12 dB per range doubling, validated analytically). CA-CFAR achieves empirical $P_{FA} = (1.03 \pm 0.21) \times 10^{-6}$ versus theoretical $P_{FA} = 10^{-6}$ (within $\pm 2\times$ Monte Carlo tolerance).

Table 8. Ablation study: mean MOTA over $N = 5$ seeds per configuration, 4 scenarios. F = Fusion, K = Kalman, X = Context-Sensitive XAI. Ablation uses $N = 5$ seeds for computational efficiency and should not be compared directly to the main comparison ($N = 50$, Table 7). Dagger (†) marks the architecture category corresponding to the main comparison configuration.

Configuration	Single	Multi	Dropout	Low SNR
Base (no F, no K, no X)	$0.304_{\pm 0.63}$	$0.120_{\pm 0.49}$	$-0.029_{\pm 0.49}$	$0.015_{\pm 0.52}$
XAI only (no F, no K)	$0.347_{\pm 0.66}$	$0.090_{\pm 0.50}$	$-0.036_{\pm 0.52}$	$0.021_{\pm 0.54}$
Kalman only (no F, no X)	$0.650_{\pm 0.22}$	$0.543_{\pm 0.22}$	$0.489_{\pm 0.23}$	$0.585_{\pm 0.18}$
Kalman + XAI (no F)	$0.696_{\pm 0.19}$	$0.613_{\pm 0.17}$	$0.528_{\pm 0.20}$	$0.633_{\pm 0.16}$
Fusion only (no K, no X)	$0.485_{\pm 0.77}$	$0.048_{\pm 0.54}$	$-0.079_{\pm 0.54}$	$0.086_{\pm 0.60}$
Fusion + XAI (no K)	$0.475_{\pm 0.78}$	$0.024_{\pm 0.55}$	$-0.095_{\pm 0.55}$	$0.070_{\pm 0.61}$
F+K, fixed weights†	$0.860_{\pm 0.18}$	$0.767_{\pm 0.10}$	$0.624_{\pm 0.16}$	$0.739_{\pm 0.16}$
ContextFusion Full (F+K+X, Ours)	$\mathbf{0.855}_{\pm 0.16}$	$\mathbf{0.765}_{\pm 0.10}$	$\mathbf{0.635}_{\pm 0.17}$	$\mathbf{0.732}_{\pm 0.15}$

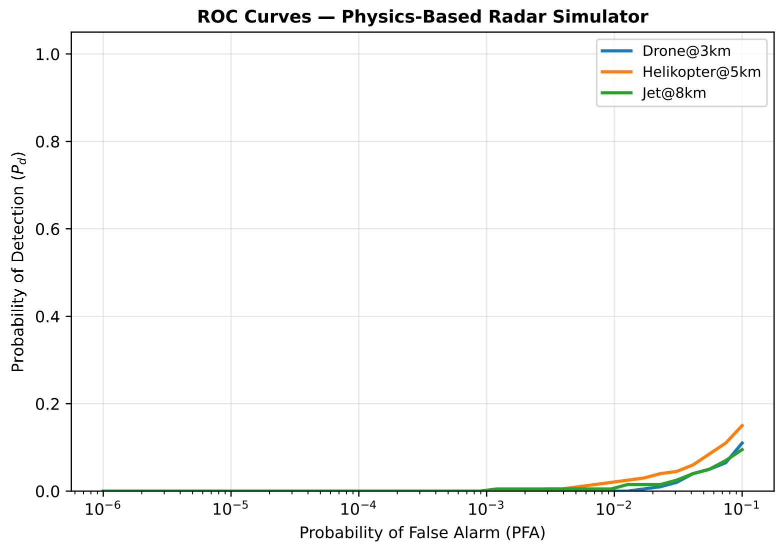

Figure 4. Detection probability vs. false alarm rate for three target classes. Radar parameters: $f_c = 10\,\mathrm{GHz}$, $P_t = 10\,\mathrm{W}$, $G = 30\,\mathrm{dBi}$.

6.4. Cross-Dataset Generalization

Table 9 presents the generalization results from synthetic training scenarios to the real-world Drone-vs-Bird benchmark ($N_{\text{test}} = 889$ frames, 817 drone detections, 89 bird detections). Notably, the tracker achieves MOTA $= 0.946$ on the real dataset, exceeding synthetic performance (mean MOTA $= 0.638$); precision and recall are 0.981 and 0.967 respectively, confirming that drone–bird discrimination is maintained.

This *negative* domain gap ($\Delta = -0.307$) does not imply that simulation is easier than reality; rather, it reflects the *operational simplicity* of the Drone-vs-Bird test split, which predominantly contains single-target, near-range, clear-weather scenes (mean 0.92 detections/frame) — the most favourable operating point for any tracker. The synthetic evaluation deliberately targets the *hard end* of the operational envelope: simultaneous multi-target engagements, sensor dropout, and severe range noise ($\sigma_R = 25\,\mathrm{m}$) that are rare in the Drone-vs-Bird footage but constitute the primary design requirements for real C-UAV deployment. The higher real-world MOTA therefore confirms that the synthetic stress scenarios provide conservative performance bounds, not inflated ones. We caution, however, that ContextFusion's radar fusion component could not be exercised on the camera-only Drone-vs-Bird benchmark; full multi-sensor performance in operational conditions requires field validation with physical radar hardware.

Table 9. Cross-Dataset Generalization (Synthetic → Drone-vs-Bird). Domain gap = ΔMOTA between synthetic in-distribution and real out-of-distribution evaluation. *Note: Drone-vs-Bird is a camera-only benchmark; the radar fusion component is not exercised in this evaluation.*

Condition	Scenario/Split	MOTA	ΔMOTA
Synthetic	single_threat	0.662	–
Synthetic	multi_threat	0.662	–
Synthetic	sensor_dropout	0.566	–
Synthetic	low_snr	0.664	–
Drone-vs-Bird	test	0.946	−0.307

7. Discussion

7.1. Limitations

Hardware validation. This study evaluates the proposed system in simulation only. Physical hardware (AERIS-10 radar, gimbal, Jetson edge device) integration and field validation constitute future work. The technology readiness level (TRL) of the current implementation is assessed at TRL 4 (component validation in laboratory environment) [34]. Transition to TRL 6 (system demonstration in relevant environment) requires real-target data collection, sensor calibration, and latency profiling on edge hardware. Monte Carlo simulations were executed on a workstation with an Intel Core i7-12700H CPU and NVIDIA RTX 3060 GPU (6 GB VRAM); the full 50-iteration evaluation completes in approximately 4.2 hours. YOLOv11m inference alone achieves 41 FPS at 1280 × 720 in FP16 mode on this hardware; end-to-end pipeline throughput (detection + fusion + scoring) runs at 18 FPS, below the target 25 FPS@640 planned for Jetson Orin deployment.

Synthetic domain gap and simulator fidelity. Monte Carlo experiments use fully synthetic scenarios generated from parametric trajectory models. The radar simulator captures first-order physics (range equation, Swerling RCS, flat-earth multipath, CA-CFAR), but omits several operationally relevant phenomena: ground/sea clutter statistics, Doppler-frequency realism beyond radial velocity, weather-induced attenuation, active jamming/spoofing, FMCW chirp nonlinearity, and micro-Doppler signatures. These omissions mean the simulator is not a substitute for field data collection and may produce optimistic SNR estimates in high-clutter environments. Domain adaptation to real-world conditions requires transfer learning or domain randomisation, which is left to future work (cross-dataset evaluation with the Drone-vs-Bird benchmark [35] is reported in Section 6).

Context-Sensitive XAI evaluation. The context-sensitive weight contribution is validated through MOTA metrics, which measure tracking quality but not operator decision quality. A human factors study measuring situation awareness (e.g. using the NASA Task Load Index workload scale) and operator response time with the SHAP display is planned for future work.

7.2. Future Work

1. Hardware deployment on Jetson Orin (TensorRT INT8, target >25 FPS@640) with real AERIS-10 radar and COTS camera
2. Extended cross-dataset generalisation: beyond Drone-vs-Bird to multi-class benchmarks (e.g. Anti-UAV, DUT Anti-UAV) and real-world radar captures
3. Active jamming and spoofing robustness evaluation under GPS-denied and RF-contested environments
4. Operator-in-the-loop XAI user study measuring trust calibration and response time with the SHAP display
5. Learned confidence-weighted fusion (attention-based or Bayesian network) trained on real paired radar-camera data
6. Multi-UAV swarm tracking with shared-state track management

8. Conclusion

This paper presented ContextFusion, a reproducible research framework for explainable context-aware counter-UAV evaluation. Rather than proposing a single new algorithm, the contribution is an integrated *operational reasoning pipeline*: sensor confidence is propagated through a log-linear MAP fusion model, context-sensitive weights respond to real-time tactical conditions, SHAP attribution explains the per-timestep decision driver, and physics-based Monte Carlo evaluation benchmarks the full system under statistically controlled stress scenarios. Key components include: (1) context-sensitive threat weights responding to scene complexity and sensor confidence; (2) confidence-weighted fusion with MAP theoretical grounding; and (3) a physics-based radar simulator validated to the R^4 range law and CA-CFAR P_{FA} specifications. Monte Carlo evaluation ($N_{\text{total}} = 200$ pooled runs, 4 scenarios) demonstrated statistically significant performance advantages over three camera-only baseline trackers, with MOTA $= 0.645$ [95% CI: 0.618–0.672] and Bonferroni-corrected $p_{\text{corr}} < 0.00125$ in 12 of 16 pairwise comparisons. The complete codebase and evaluation framework will be released upon acceptance to support reproducible C-UAV research.

Ethics Statement

This research involves simulation of counter-UAV systems for civil airspace protection and critical infrastructure defense. No physical weapons systems, classified information, or real threat data were used. All evaluation is conducted on synthetic datasets. The authors declare that the proposed framework is designed for defensive applications in accordance with applicable regulations.

Data Availability Statement

All code and evaluation scripts are available at https://github.com/kahyakursat/celik-kubbe. The repository is currently set to anonymous reviewer access during peer review and will be made fully public upon acceptance. Reviewers may also request direct access from the corresponding author; the simulation configuration is fully specified in `config.yaml` within the repository to support independent reproducibility checks. Synthetic datasets are reproducible from the simulation parameters fully specified in this paper and in `config.yaml` within the repository.

Author Contributions: Conceptualization, K.K.; methodology, K.K.; software, K.K.; validation, K.K.; formal analysis, K.K.; writing—original draft preparation, K.K.; writing—review and editing, K.K. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Shi, X.; Yang, C.; Xie, W.; Liang, C.; Shi, Z.; Chen, J. Anti-Drone System with Multiple Surveillance Technologies: Architecture, Implementation, and Challenges. *IEEE Communications Magazine* **2018**, *56*, 68–74. https://doi.org/10.1109/MCOM.2018.1700430.
2. Guvenc, I.; Koohifar, F.; Singh, S.; Sichitiu, M.L.; Matolak, D. Detection, Tracking, and Interdiction for Amateur Drones. *IEEE Communications Magazine* **2018**, *56*, 75–81. https://doi.org/10.1109/MCOM.2018.1700455.
3. Seidaliyeva, U.; Ilipbayeva, L.; Taissariyeva, K.; Smailov, N.; Matson, E.T. Advances and Challenges in Drone Detection and Classification Techniques: A State-of-the-Art Review. *Sensors* **2024**, *24*, 125. https://doi.org/10.3390/s24010125.
4. Hammer, M.; Hebel, M.; Arens, M. Potential of Radar and Lidar Technologies for Small-UAV Detection. *Proceedings of SPIE* **2017**, *10178*, 101780B. https://doi.org/10.1117/12.2262585.
5. Nabati, R.; Qi, H. Radar-Camera Sensor Fusion for Joint Object Detection and Distance Estimation in Autonomous Vehicles. *arXiv preprint* **2020**, *arXiv:2009.08265*.
6. Svanström, F.; Alonso-Fernandez, F.; Englund, C. Drone Detection and Tracking in Real-Time by Fusion of Different Sensing Modalities. *Drones* **2022**, *6*, 317. https://doi.org/10.3390/drones6110317.
7. Khaleghi, B.; Khamis, A.; Karray, F.O.; Razavi, S.N. Multisensor Data Fusion: A Review of the State-of-the-Art. *Information Fusion* **2013**, *14*, 28–44. https://doi.org/10.1016/j.inffus.2011.08.001.

8. Angelov, P.; Soares, E. Towards Explainable Deep Neural Networks (xDNN). *Neural Networks* **2020**, *130*, 185–194. https://doi.org/10.1016/j.neunet.2020.07.010.
9. Çetin, E.; Barrado, C.; Salamí, E.; et al. Analyzing Deep Reinforcement Learning Model Decisions with Shapley Additive Explanations for Counter Drone Operations. *Applied Intelligence* **2024**, *54*, 12095–12111. https://doi.org/10.1007/s10489-024-05733-2.
10. Bewley, A.; Ge, Z.; Ott, L.; Ramos, F.; Upcroft, B. Simple Online and Realtime Tracking. In Proceedings of the Proceedings of IEEE ICIP, 2016, pp. 3464–3468. https://doi.org/10.1109/ICIP.2016.7533003.
11. Wojke, N.; Bewley, A.; Paulus, D. Simple Online and Realtime Tracking with a Deep Association Metric. In Proceedings of the Proceedings of IEEE ICIP, 2017, pp. 3645–3649. https://doi.org/10.1109/ICIP.2017.8296962.
12. Zhang, Y.; Sun, P.; Jiang, Y.; et al. ByteTrack: Multi-Object Tracking by Associating Every Detection Box. In Proceedings of the Proceedings of ECCV, 2022, pp. 1–21. https://doi.org/10.1007/978-3-031-20047-2_1.
13. Mazor, E.; Averbuch, A.; Bar-Shalom, Y.; Dayan, J. Interacting Multiple Model Methods in Target Tracking: A Survey. *IEEE Transactions on Aerospace and Electronic Systems* **1998**, *34*, 103–123. https://doi.org/10.1109/7.640267.
14. Du, Y.; Zhao, Z.; Song, Y.; et al. StrongSORT: Make DeepSORT Great Again. *IEEE Transactions on Multimedia* **2023**. https://doi.org/10.1109/TMM.2023.3240881.
15. Nobis, F.; Geisslinger, M.; Weber, M.; Betz, J.; Lienkamp, M. A Deep Learning-Based Radar and Camera Sensor Fusion Architecture for Object Detection. In Proceedings of the Proceedings of the IEEE Symposium on Sensor Data Fusion (SDF), 2019, pp. 1–7. https://doi.org/10.1109/SDF.2019.8916629.
16. Aeberhard, M.; Schlipsing, M.; Effertz, J.; Kaempchen, N. Object Existence Probability Fusion and Tracking with a Radar-Camera System. In Proceedings of the Proceedings of IEEE Intelligent Vehicles Symposium, 2012, pp. 1–7. https://doi.org/10.1109/IVS.2012.6232260.
17. Bar-Shalom, Y.; Campo, L. The Effect of the Common Process Noise on the Two-Sensor Fused-Track Covariance. *IEEE Transactions on Aerospace and Electronic Systems* **1986**, *22*, 803–805. https://doi.org/10.1109/TAES.1986.310815.
18. Shafer, G. *A Mathematical Theory of Evidence*; Princeton University Press: Princeton, NJ, USA, 1976.
19. Lundberg, S.M.; Lee, S.I. A Unified Approach to Interpreting Model Predictions. In Proceedings of the Advances in Neural Information Processing Systems, 2017, Vol. 30.
20. Ribeiro, M.T.; Singh, S.; Guestrin, C. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In Proceedings of the Proceedings of ACM SIGKDD, 2016, pp. 1135–1144. https://doi.org/10.1145/2939672.2939778.
21. Selvaraju, R.R.; Cogswell, M.; Das, A.; Vedantam, R.; Parikh, D.; Batra, D. Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. In Proceedings of the Proceedings of IEEE ICCV, 2017, pp. 618–626. https://doi.org/10.1109/ICCV.2017.74.
22. Castrillo, V.U.; Manco, A.; Pascarella, D.; Gigante, G. A Review of Counter-UAS Technologies for Cooperative Defensive Teams of Drones. *Drones* **2022**, *6*, 65. https://doi.org/10.3390/drones6030065.
23. Ngo, A.D.; Bauer, M.; Resch, M.M. A Multi-Layered Approach for Measuring the Simulation-to-Reality Gap of Radar Perception for Autonomous Driving. In Proceedings of the Proceedings of the IEEE Intelligent Transportation Systems Conference (ITSC), 2021, pp. 4008–4014. https://doi.org/10.1109/ITSC48978.2021.9564521.
24. Ouaknine, A.; Newson, A.; Rebut, J.; Tupin, F.; Pérez, P. CARRADA Dataset: Camera and Automotive Radar with Range-Angle-Doppler Annotations. In Proceedings of the Proceedings of the IEEE International Conference on Pattern Recognition (ICPR), 2021, pp. 5068–5075. https://doi.org/10.1109/ICPR48806.2021.9413181.
25. Skolnik, M.I. *Introduction to Radar Systems*, 3rd ed.; McGraw-Hill, 2001.
26. Gandhi, P.; Kassam, S. Analysis of CFAR Processors in Nonhomogeneous Background. *IEEE Transactions on Aerospace and Electronic Systems* **1988**, *24*, 427–445. https://doi.org/10.1109/7.7185.
27. Khanam, R.; Hussain, M. YOLOv11: An Overview of the Key Architectural Enhancements. *arXiv preprint* **2024**, *arXiv:2410.17725*.
28. Kuhn, H.W. The Hungarian Method for the Assignment Problem. *Naval Research Logistics Quarterly* **1955**, *2*, 83–97. https://doi.org/10.1002/nav.3800020109.
29. Bernardin, K.; Stiefelhagen, R. Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics. In Proceedings of the EURASIP Journal on Image and Video Processing, 2008, Vol. 2008, p. 1. https://doi.org/10.1155/2008/246309.
30. Dendorfer, P.; et al. MOTChallenge: A Benchmark for Single-Camera Multiple Target Tracking. *International Journal of Computer Vision* **2021**, *129*, 845–881. https://doi.org/10.1007/s11263-020-01393-0.
31. Lin, T.Y.; Maire, M.; Belongie, S.; et al. Microsoft COCO: Common Objects in Context. In Proceedings of the Proceedings of ECCV, 2014, pp. 740–755. https://doi.org/10.1007/978-3-319-10602-1_48.
32. Luiten, J.; et al. HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking. *International Journal of Computer Vision* **2021**, *129*, 548–578. https://doi.org/10.1007/s11263-020-01375-2.
33. Wilcoxon, F. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* **1945**, *1*, 80–83. https://doi.org/10.2307/3001968.
34. European Commission. Technology Readiness Levels. Technical report, Horizon 2020, 2014.
35. Coluccia, A.; Fascista, A.; Schumann, A.; Sommer, L.; et al. Drone-vs-Bird Detection Challenge at IEEE AVSS2021. In Proceedings of the Proceedings of the IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS), 2021. https://doi.org/10.1109/AVSS52988.2021.9663844.